HW #2. DIKW

Note: Answer the following questions in Chinese except that you are a foreign student.

1. Read the article "Business Analytics Professional or Data Scientist?" from INFORMS Analytics: https://pubsonline.informs.org/do/10.1287/LYTX.2020.06.07/full/.

   Try to analyze your own skills using the online spreadsheet https://docs.google.com/forms/d/e/1FAIpQLSfefZGlAFQxGGoncm-Xdb68y46XLikGuzn2VHxIazZLJi1Cig/viewform

   built from the article to identify if yours aligns with that of a business analytics professional or data scientist.

   You should complete this analysis according to the two different cases: you possess the skill AND you are interested in acquiring the skill.

   Answer the following analytical questions.

   (1) Identify your alignment.
   (2) Design a good visual presentation for your result using excel or Google Docs.
   (3) Identify the well-defined problems or opportunities for your future career based on your analysis.

2. The DIKW (Data, Information, Knowledge, Wisdom) model serves as a structured guide for organizations to transform data into strategic decisions. In this assignment, you will apply the DIKW model to a challenge or opportunity you have encountered in your professional experience or through a case study from news or professional media. You will analyze the scenario using the DIKW framework and demonstrate how it informs decision-making.

   (1) Explain the difference between Knowledge and Wisdom within the context of organizational decision-making. Provide an example to

illustrate the distinction and the unique role each plays in the process.

(2) Applying DIKW Analysis to a Specific Challenge or Opportunity for Your Decision-Making Task:

Data (D): Identify the raw data relevant to this scenario.

Information (I): Describe how this data is contextualized to become meaningful information.

Knowledge (K): Discuss the patterns or relationships discerned from the information that contribute to a deeper understanding.

Wisdom (W): Articulate the principled understanding or ethical considerations that stem from the knowledge.

Insight (I): Elucidate on how insight is gleaned from the analysis.

(3) Explain how you apply insight to evolve knowledge into wisdom within the scenario. Discuss the cognitive processes or analytical methods that facilitate this transformation.

(4) Propose a set of strategic decisions or actions to tackle the identified challenge. Explain how each decision is grounded in the data, information, knowledge, and wisdom acquired, including the application of insights to guide these decisions.

3. Navigate to the Amazon Science Blog at https://www.amazon.science/blog/, which showcases "The latest news from Amazon's science community."

   Select and read any blog article published within the past one year that catches your interest. Additionally, review the "Related Publications" listed on the left-hand side of the webpage to gain insights into the methods applied in the research.

   (1) Business Problem: Describe the specific business issue the article addresses.

   (2) Amazon's Solution: Explain how Amazon approached and solved this problem. Detail the technologies or methodologies employed.

   (3) Business Value: Identify the direct business benefits that result from Amazon's solution. Consider aspects such as cost savings, efficiency improvements, customer satisfaction, etc.

   (4) Potential Applications: Discuss other possible applications for the method adopted by Amazon. Consider different industries or scenarios where this approach could be beneficial.

4. Reflecting on the case study discussed in class

(https://news.cnyes.com/news/id/5332964) , consider how Descriptive, Predictive, and Prescriptive Analytics can be applied in your own subject area or industry of interest. Your goal is to conceptualize a real-world analytics use case, not to implement it, but to develop a clear and structured idea.

Identify a business problem or opportunity within your chosen field and think about how data-driven insights could help address it.

In your response, clearly outline the following:

(1) Business Problem:

- Describe the specific issue or opportunity that your analytics use case addresses.
- Why is this problem important? Who are the key stakeholders?

(2) Analytics Approach:

- Descriptive Analytics: What historical data would be collected? How would it help in understanding past trends or behaviors?
- Predictive Analytics: What models or techniques could be used to forecast future trends or outcomes?
- Prescriptive Analytics: Based on the predictions, what actions or recommendations could be made to optimize decision-making?

(3) Business Value:

- Identify the direct benefits of implementing this analytics-driven solution.
- Consider aspects such as cost reduction, efficiency improvements, risk mitigation, customer satisfaction, or competitive advantage.